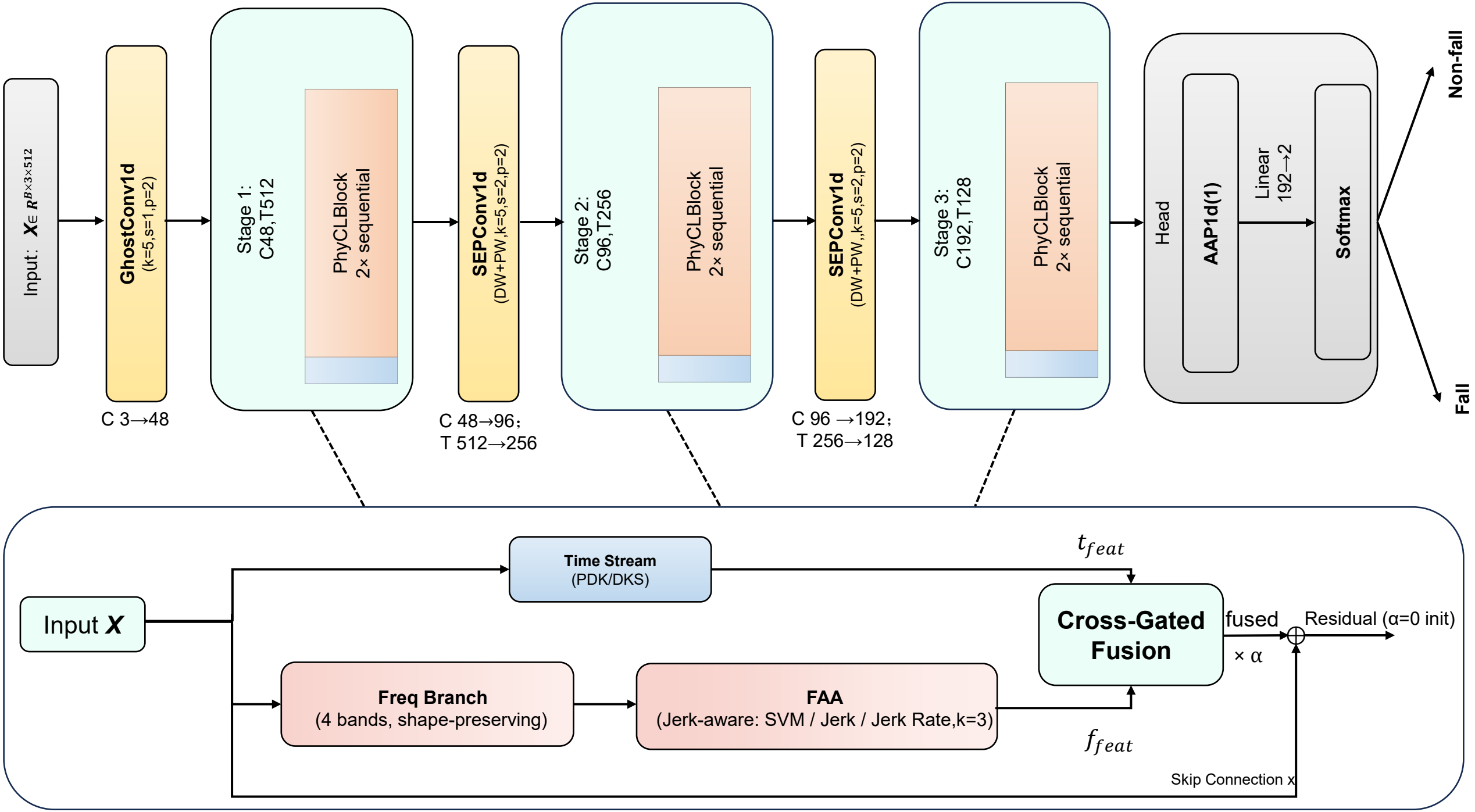
Input: $X \in R^{B\times3\times512}$
GhostConv1d
(k=5,s=1,p=2)
C 3→48
Stage 1:
C48,T512
PhyCLBlock
2× sequential
SEPConv1d
(DW+PW,k=5,s=2,p=2)
C 48→96;
T 512→256
Stage 2:
C96,T256
PhyCLBlock
2× sequential
SEPConv1d
(DW+PW,,k=5,s=2,p=2)
C 96 →192;
T 256→128
Stage 3:
C192,T128
PhyCLBlock
2× sequential
Head
AAP1d(1)
Linear
192→2
Softmax
Non-fall
Fall
Input X
Time Stream
(PDK/DKS)
t_{feat}
Cross-Gated
Fusion
fused
× α
Residual (α=0 init)
Freq Branch
(4 bands, shape-preserving)
FAA
(Jerk-aware: SVM / Jerk / Jerk Rate,k=3)
f_{feat}
Skip Connection x